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Dynamic Vibration Analysis of Perforated Non-Homogeneous Nanobeams on Variable Elastic Foundations for Multiphysics Nano-Engineering Applications

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Abstract: The dynamic behavior of nanoscale beam structures is strongly influenced by material gradation, geometric discontinuities, foundation characteristics, and small-scale effects, all of which play critical roles in the performance of advanced multiphysics nano-engineering systems. In the present study, the free vibration characteristics of a perforated non-homogeneous nanobeam resting on a variable elastic foundation were investigated under sliding-end boundary conditions. Particular attention was devoted to applications involving resonators of microelectromechanical and nanoelectromechanical systems, nano-sensors, smart structural components, and coupled electromechanical nano-devices, where precise control of dynamic response is essential. Spatial variations in Young's modulus and material density were incorporated to represent non-homogeneous material properties, while perforation effects were introduced through modified geometric and mechanical characteristics. Size-dependent nanoscale behavior was captured using Eringen's nonlocal elasticity theory. Based on Euler-Bernoulli beam theory, the governing differential equation of motion was derived. The resulting eigenvalue problem was solved using the Galerkin method in conjunction with shifted Legendre polynomial admissible functions, enabling high numerical stability, rapid convergence, and computational efficiency. Validation of the proposed formulation was performed through comparisons with available benchmark results reported in the literature, and additional convergence studies were conducted. Investigations were carried out to evaluate the effects of perforation characteristics, non-homogeneity parameters, and spatially varying foundation stiffness on the natural frequencies and mode shapes of the nanobeam. It was demonstrated that significant alterations in dynamic response may be induced by the combined interaction of material gradation, perforation geometry, and foundation variability. The developed model provides an efficient and reliable computational framework for the dynamic characterization of perforated nanoscale structures and offers valuable insights for the design, optimization, and vibration control of next-generation multifunctional nano-engineering systems operating under coupled multiphysics environments.

Keywords: Perforated nanobeam; Vibration analysis; Legendre polynomial; Galerkin method; Non-homogeneity; Elastic foundation; Microelectromechanical and nanoelectromechanical systems; Multiphysics nano-systems

1 Introduction

Solid structural elements have long been fundamental in engineering due to their ability to sustain loads and provide stability in various applications. Among these, beams are one of the most widely used components, serving as primary load-carrying members in civil, mechanical, and aerospace structures. With the advancement of technology, the concept of beams has been extended to the nanoscale, leading to the development of nanobeams, which are essential in advanced microelectromechanical and nanoelectromechanical systems owing to their lightweight characteristics, high sensitivity, and superior dynamic performance. They are widely utilized in nano-resonators, biosensors [1], smart actuators, precision instruments, energy harvesting systems, and coupled electromechanical nano-devices.

In recent years, considerable attention has been devoted to nanoscale structures operating under coupled multiphysics environments involving mechanical, thermal, electrical, magnetic, and elastic effects. Accurate prediction of nanobeam vibration behavior considering small-scale effects, material gradation, perforation, and substrate interactions is therefore essential for the reliable design of advanced nano-engineering systems. In modern engineering design, beams are no longer restricted to solid configurations. Instead, porous and perforated beams [2–4] have gained considerable attention, where material is systematically removed in the form of pores or openings. Such designs offer advantages like reduced weight, improved material efficiency, and tunable mechanical properties. However, the presence of perforations significantly alters the stiffness and mass distribution, making the structural behavior more complex.

These structures play an important role in the design and development of advanced resonators of microelectromechanical and nanoelectromechanical systems, nano-sensors, smart structural components, and coupled electromechanical systems where accurate vibration control and frequency tuning are critically important [5, 6]. In addition, the incorporation of material non-homogeneity offers greater flexibility in controlling the spatial distribution of stiffness and mass density along the nanobeam, leading to optimized modal characteristics and improved structural functionality [7]. Moreover, the consideration of variable elastic foundation parameters allows the proposed model to realistically represent substrate interactions, support flexibility, and environmental constraints commonly encountered in practical nano-engineering applications [8]. The combined influence of nonlocal effects, perforation distribution, material gradation, and foundation variability provides a more comprehensive and physically meaningful representation of nanoscale structural behavior for the analysis and design of advanced nanobeam systems operating under coupled multiphysics conditions.

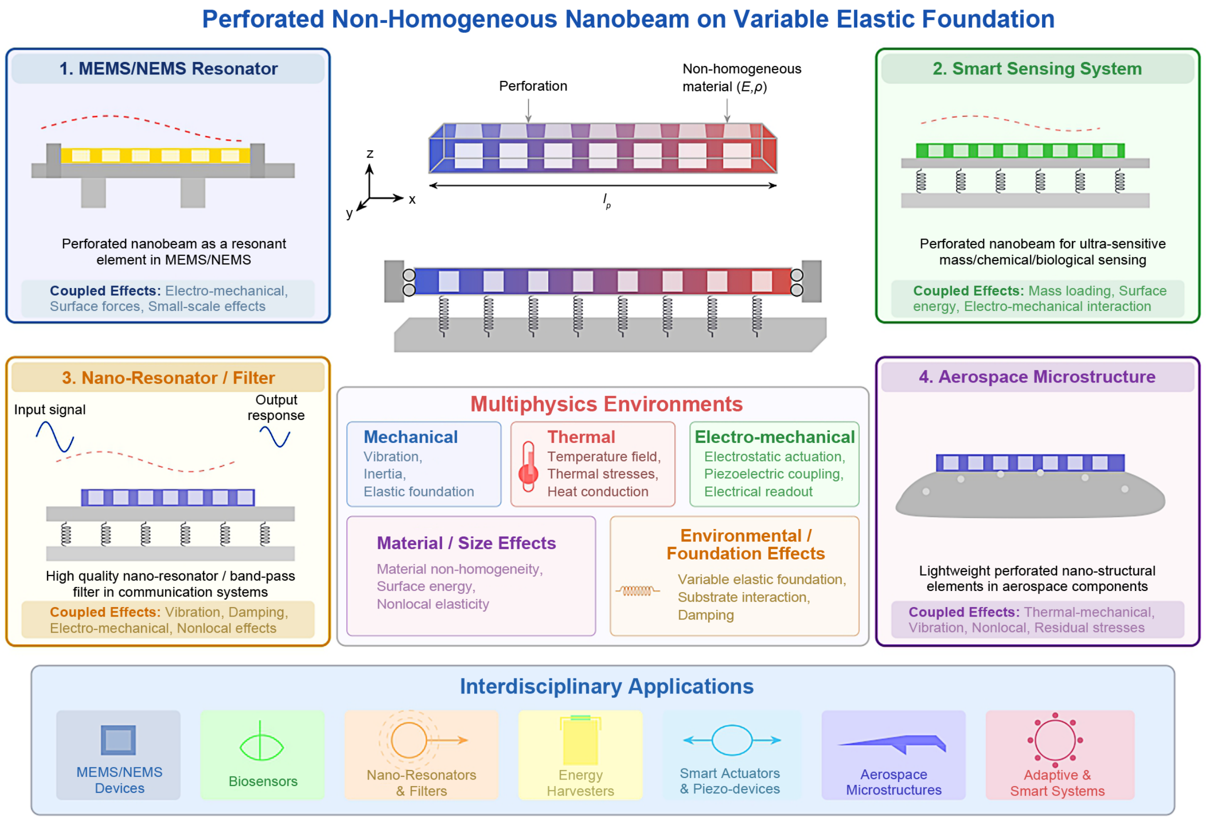


Figure 1. Schematic representation of engineering applications of a perforated nanobeam

Note: MEMS/NEMS = Microelectromechanical and nanoelectromechanical systems.

Luschi and Pieri [4, 9] proposed an analytical formulation for perforated beams by deriving equivalent bending and shear stiffness. Their model was applied to evaluate the resonance frequencies of clamped-clamped beams in resonant microelectromechanical systems. Hong et al. [10] investigated the natural frequencies and mode shapes of simply supported perforated beams filled with ideal fluids using the Rayleigh-Ritz method and demonstrated the applicability of the developed model to piezoelectric inkjet printer head structures. Zulkefli et al. [11] performed finite element simulations of suspended double-clamped graphene beam-based nanoelectromechanical switches with standard and perforated configurations and demonstrated that the introduction of perforations significantly reduced the von Mises stress and improved the mechanical reliability and electrostatic performance of graphene-based nanoelectromechanical switch devices. Şimşek [12] analyzed the nonlinear free vibration of functionally graded nanobeams using a size-dependent nonlocal strain gradient model based on Euler-Bernoulli theory with von Kármán nonlinearity for immovable ends. Yu et al. [13] studied perforated thin-walled beams with web openings and showed that these holes change the stress distribution and can lead to distortional buckling under transverse loads. Kafkas et al. [14] investigated the thermal vibration behavior of perforated nanobeams resting on a Winkler elastic foundation by incorporating nonlocal and strain gradient effects under rigid and deformable boundary conditions, and demonstrated the significant influence of elastic foundation stiffness, thermal effects, perforation parameters, and small-scale effects on the vibration characteristics of nanoscale beam structures. Penna et al. [15] investigated the buckling behavior of functionally graded Euler-Bernoulli nanobeams with different cross-sectional shapes using a surface stress-based elasticity model. Güçlü [16] investigated the nonlocal free vibration behavior of perforated nanobeams resting on a Kerr-type elastic foundation using Eringen's nonlocal elasticity theory. Their study incorporated the three-parameter Kerr foundation model to account for shear continuity and realistic surface interactions while capturing small-scale effects. Hong et al. [17] studied a perforated beam with piezoelectric patches to achieve tunable topological behavior, analyzing its wave propagation using a Timoshenko beam model. Gartia et al. [18] developed a model to study the free vibration of axially graded tapered nanobeams with square perforations, highlighting the combined effects of material gradation and geometry. Abdelrahman et al. [6] developed a size-dependent mathematical model and analytical solution to investigate the buckling behavior of piezoelectric sandwich perforated nanobeams resting on an elastic foundation. Their model incorporates flexoelectric effects through the electric enthalpy energy formulation to capture size-dependent behavior accurately.

Although significant progress has been made in perforated and functionally graded nanobeams, the combined effects of nonlocal elasticity, material non-homogeneity, perforation, and variable elastic foundation under sliding-end boundary conditions remain unexplored. Motivated by the above research gap, the present work investigates the free vibration characteristics and mode shape deflections of a non-homogeneous perforated nanobeam resting on a variable elastic foundation under sliding-end boundary conditions. The remaining study is organized as follows: Section 2 presents the mathematical representation of the perforated nanobeam. Section 3 contains the development of this present model. Section 4 presents the solution procedure using the Galerkin method with Legendre polynomials, and Section 5 provides the validation of this present model. Sections 6 and 7 contain the new findings and the overall conclusion of this research. Figure 1 shows a schematic representation of engineering applications of a perforated nanobeam.

2 Mathematical Representation of a Perforated Nanobeam

A perforated nanobeam is a structural element in which a regular pattern of holes is introduced to modify its mechanical behavior. These perforations can take different shapes, such as circular or rectangular, depending on design requirements. In this work, square-shaped holes, arranged in a uniform grid pattern, are considered. The presence of these perforations reduces the overall weight of the nanobeam while influencing its stiffness, strength, and vibration characteristics.

In the present analysis, the nanobeam is treated as a non-homogeneous system in which material properties vary along its length. In particular, both Young's modulus and mass density are considered as functions of the axial coordinate x. A polynomial-based representation is adopted to describe this spatial variation, while the change in density is assumed to be small. Such non-uniformity in material properties affects both the stiffness and inertia distribution of the nanobeam. In this work, the spatial dependence of Young's modulu $E(x)$ and density $\rho(x)$ are expressed using quadratic functions of x, as shown below:

$$E(x) = E_s \left(1 + \alpha \frac{x}{l_p} + \beta \left(\frac{x}{l_p}\right)^2\right) \tag{1}$$

$$\rho(x) = \rho_s \left(1 + \gamma \frac{x}{l_p} + \delta \left(\frac{x}{l_p}\right)^2\right) \tag{2}$$

where, E_s and ρ_s represent Young's modulus and density of the solid beam material, respectively. The coefficients α, β, γ, and δ act as controlling parameters that characterize the variation of stiffness and mass distribution along the nanobeam.

This section shows the modified equivalent geometric models. Because of non-homogeneity and holes present in the nanobeam, the material's shape changes, and new formulas for bending stiffness, mass per unit length, and rotating inertia are given in this study.

A perforated nanobeam geometry is shown in Figure 2. As presented, the perforated nanobeam has length l_p, width w_p and height h_p. The nanobeam has a pattern of square holes of spatial period S_θ, the period length P_θ, side of the square holes $S_\theta - P_\theta$ and H_n is the number of holes along the section. The ratio of the period length P_θ to the spatial period S_θ, referred to as the filling ratio of the nanobeam, is given by [4, 19]:

$$\eta = \frac{P_\theta}{S_\theta}, 0 \leq \eta \leq 1 \tag{3}$$

Thus, the nanobeam is fully solid when the filling ratio $\eta = 1$, partially filled for $0 < \eta < 1$, and completely perforated when $\eta = 0$.

Therefore, the mathematical expression is as follows [4]:

$$\eta = \begin{cases} 0 & \text{Completely perforated (Artificial case)} \\ 0 < \eta < 1 & \text{Partiallyfilled} \\ 1 & \text{Completely filled} \end{cases}$$

Due to the combined effects of non-homogeneity and square holes in the nanobeam, the bending stiffness model is revised as [4, 19]:

$$[EI]_{eq} = [E(x)I]_s \left[\frac{\eta\left(H_n+1\right)\left(H_n^2+2H_n+\eta^2\right)}{\left(1-\eta^2+\eta^3\right)H_n^3+3\eta H_n^2+\left(3+2\eta-3\eta^2+\eta^3\right)\eta^2 H_n+\eta^3} \right] \tag{4}$$

where, $[E(x)]_s$ is the bending stiffness of the solid nanobeam, E and I are the Young's modulus and moment of inertia of the nanobeam, respectively.

The modified mass per unit length of the non-homogeneous perforated nanobeam can be expressed as [20]:

$$[\rho A]_{eq} = [\rho(x)A]_s \left[\frac{\left[1-H_n(\eta-2)\right]\eta}{H_n+\eta} \right] \tag{5}$$

where, $[\rho(x)A]_s$ represents the mass per unit length of the solid nanobeam.

The modified moment of inertia per unit length is obtained by performing integration and averaging over a strip containing η square cells, each of length l_p, expressed as [9, 19]:

$$[\rho I]_{\mathrm{eq}} = [\rho(x)I]_s \left[\frac{\eta\left[(2-\eta)H_n^3+3H_n^2-2(\eta-3)\left(\eta^2-\eta+1\right)H_n+\eta^2+1\right]}{\left(H_n+\eta\right)^3} \right] \tag{6}$$

where, $[\rho(x)I]_s$ represents the rotational inertia per unit length of the solid nanobeam.

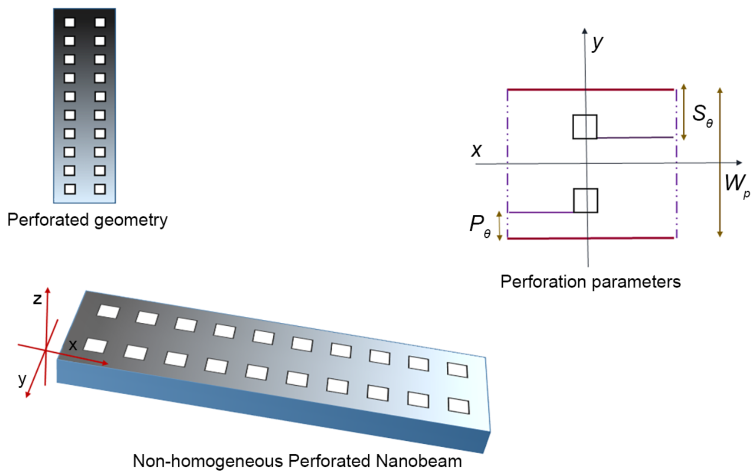


Figure 2. Geometric model of a non-homogeneous perforated nanobeam

Note: S_θ–spatial period of the perforation pattern; P_θ–period length; w_p–width of the nanobeam; x–longitudinal coordinate; y–transverse coordinate along the width; z–coordinate along the thickness (height).

3 Problem Development

According to the Euler-Bernoulli beam theory, the rotation of the beam's cross-section is assumed to be negligible relative to its transverse displacement. Additionally, the theory assumes that the angular distortion caused by shear deformation is insignificant when compared to the bending deformation. The displacement variables V_1, V_2 and V_3 associated with the X, Y and Z directions, in the context of Euler-Bernoulli beam theory, are expressed as [20]:

$$V_1(x,z,t) = B_p(x,t) - z\frac{\partial G_p(x,t)}{\partial x} \tag{7}$$

$$V_2(x,z,t) = 0 \tag{8}$$

$$V_3(x,z,t) = G_p(x,t) \tag{9}$$

where, axial and transverse displacements along the neutral axis are denoted by $B_p(x,t)$ and $G_p(x,t)$, respectively, and t denotes the time. The axial displacement is neglected due to its comparatively small magnitude relative to the transverse displacement and rotation.

Under the assumption of small deformations, and based on the Euler-Bernoulli beam theory, the only non-zero strain component is normal strain along the beam axis (ϵ_{xx}):

$$\epsilon_{xx} = \frac{\partial V_1(x,z,t)}{\partial x} = \frac{\partial B_p(x,t)}{\partial x} - z\frac{\partial^2 G_p(x,t)}{\partial x^2} \tag{10}$$

Based on Eringen's non-local theory, the bending moment resultant is expressed as [21, 22]:

$$M_R - (e_0 a)^2\,\frac{\partial^2 M_R}{\partial x^2} = -[EI]_{eq}\frac{\partial^2 G_p}{\partial x^2} \tag{11}$$

Transverse vibration of a non-homogeneous perforated nanobeam with a variable elastic foundation is governed by the Lagrange equation of motion, which is written as:

$$\frac{\partial^2 M_R}{\partial x^2} = S_0\frac{\partial^2 G_p}{\partial t^2} - S_2\frac{\partial^4 G_p}{\partial t^2 \partial x^2} + K_w(x)G_p \tag{12}$$

The elastic foundation is assumed to vary quadratically along the nanobeam length, and is given by:

$$K_w(x) = K_0\left(1 + g\frac{x}{l_p} + \mathrm{h}\left(\frac{x}{l_p}\right)^2\right) \tag{13}$$

where, K_0 is the reference foundation stiffness, while g and h are parameters controlling the linear and quadratic variation of the foundation stiffness along the nanobeam and $S_0 = [\rho A]_{eq}$ and $S_2 = [\rho I]_{eq}$.

Substituting Eq. (12) into Eq. (11), the governing differential equation for the non-homogeneous perforated nanobeam may be obtained as:

$$\begin{aligned} -\frac{\partial^2}{\partial x^2}\left([EI]_{eq}\frac{\partial^2 \mathrm{G}_p}{\partial x^2}\right) + (e_0 a)^2\,\frac{\partial^2}{\partial x^2}\left(S_0\frac{\partial^2 \mathrm{G}_p}{\partial t^2} - S_2\frac{\partial^4 \mathrm{G}_p}{\partial t^2\partial x^2} + K_w(x)\mathrm{G}_p\right) \\ = S_0\frac{\partial^2 \mathrm{G}_p}{\partial t^2} - S_2\frac{\partial^4 \mathrm{G}_p}{\partial t^2 \partial x^2} + K_w(x)G_p \end{aligned} \tag{14}$$

For the free vibration analysis considering the simple harmonic motion, substitution of $\mathrm{G}_p(x,t) = \widetilde{\mathrm{G}_p}(x)\cos(\omega t)$ into Eq. (14) leads to:

$$\begin{aligned} -\frac{d^2}{dx^2}\left([EI]_{eq}\frac{d^2\widetilde{\mathrm{G}}_p}{dx^2}\right) + (e_0 a)^2\,\frac{d^2}{dx^2}\left(-\omega^2 S_0\widetilde{\mathrm{G}}_p + \omega^2 S_2\frac{d^2\widetilde{\mathrm{G}}_p}{dx^2} + K_w(x)\widetilde{\mathrm{G}}_p\right) \\ = -\omega^2 S_0\widetilde{\mathrm{G}}_p + \omega^2 S_2\frac{d^2\widetilde{\mathrm{G}}_p}{dx^2} + K_w(x)\widetilde{\mathrm{G}}_p \end{aligned} \tag{15}$$

where, $\widetilde{G}_p(x)$ represents the mode shape of the non-homogeneous perforated nanobeam, and ω is the natural frequency.

3.1 Non-Dimensional Form

Non-dimensionalization transforms a differential equation into a unit-free form, allowing its results to be applied broadly and compared across different physical systems. The following set of non-dimensional terms is introduced:

$$X = \frac{x}{l_p}, \bar{G}_p(X) = \frac{\tilde{G}_p(X)}{l_p} \tag{16}$$

Hence, the differential relation becomes:

$$dX = \frac{dx}{l_p} \text{ and } \frac{d\overline{\mathrm{G}}_p(X)}{dX} = \frac{d}{dX}\left(\frac{\widetilde{\mathrm{G}}_p(x)}{l_p}\right)$$

It can be written as:

$$\frac{d\bar{G}_p(X)}{dX} = \frac{d}{dx}\left(\frac{\widetilde{G}_p(x)}{l_p}\right) \times \frac{dx}{dX} = \frac{d}{dx}\left(\frac{\widetilde{G}_p(x)}{l_p}\right) \times l_p = \frac{d\widetilde{G}_p(x)}{dx}$$

Similarly for the second-order derivative form:

$$\frac{d^2\bar{G}_p(X)}{dX^2} = l_p \times \frac{d\widetilde{G}_p(x)}{dx}$$

For the operator in differentiation form:

$$\frac{d}{dX} = \frac{d}{dx} \cdot \frac{dx}{dX} \Rightarrow \frac{d}{dX} = l_p \times \frac{d}{dx}$$

After substituting the above non-dimensional terms into the above Eq. (15), the dimensionless differential equation is obtained as:

$$\begin{aligned}
-\frac{1}{l_p^2} \times \frac{d^2}{dX^2}\left([EI]_{eq} \times \frac{1}{l_p} \times \frac{d^2\overline{\mathrm{G}}_p}{dX^2}\right) + (e_0 a)^2 \times \frac{1}{l_p^2}\frac{d^2}{dX^2}\left(-\omega^2 S_0 l_p \overline{\mathrm{G}}_p + \omega^2 S_2 \frac{1}{l_p}\frac{d^2\overline{\mathrm{G}}_p}{dX^2} + K_w l_p \overline{\mathrm{G}}_p\right) \\
= -\omega^2 S_0 l_p \overline{\mathrm{G}}_p + \omega^2 S_2 \frac{1}{l_p}\frac{d^2\overline{\mathrm{G}}_p}{dX^2} + K_w l_p \overline{\mathrm{G}}_p
\end{aligned}$$

After simplification, it is expressed as:

$$\begin{aligned}
\frac{d^2}{dX^2}\left([EI]_{eq}^* \frac{d^2\bar{G}_p}{dX^2}\right) + \lambda^2\xi^2 \frac{d^2}{dX^2}\left(S_0^*\bar{G}_p - S_2^* \frac{h_p^2}{12 \times l_p^2}\frac{d^2\bar{G}_p}{dX^2}\right) \\
-K_w^*\xi^2 \frac{d^2}{dX^2}\left(\left(1 + gX + hX^2\right)\bar{G}_p\right) = \lambda^2\left(S_0^*\bar{G}_p - S_2^* \frac{h_p^2}{12 \times l_p^2}\frac{d^2\bar{G}_p}{dX^2}\right) - K_w^*\left(\left(1 + gX + hX^2\right)\bar{G}_p\right)
\end{aligned} \tag{17}$$

where,

$$\lambda^2 = \frac{\omega^2 l_p^4 \rho_s A}{E_s I}, \xi^2 = \left(\frac{e_0 a}{l_p}\right)^2, K_w^* = \frac{K_0 l_p^4}{E_s I},$$

$$\left[EI_{eq}^* = \left[1 + \alpha X + \beta X^2\right]\left[\frac{\eta\left(H_n + 1\right)\left(H_n^2 + 2H_n + \eta^2\right)}{\left(1 - \eta^2 + \eta^3\right)H_n^3 + 3\eta H_n^2 + \left(3 + 2\eta - 3\eta^2 + \eta^3\right)\eta^2 H_n + \eta^3}\right]\right.$$

$$S_0^* = [\rho A]_{eq}^* = \left[1 + \gamma X + \delta X^2\right]\left[\frac{\left[1 - H_n(\eta - 2)\right]\eta}{H_n + \eta}\right]$$

$$S_2^* = [\rho I]_{eq}^* = \left[1 + \gamma X + \delta X^2\right]\left[\frac{\eta\left[(2 - \eta)H_n^3 + 3H_n^2 - 2(\eta - 3)\left(\eta^2 - \eta + 1\right)H_n + \eta^2 + 1\right]}{\left(H_n + \eta\right)^3}\right].$$

4 Solution Procedure

The present study employs the Euler-Bernoulli beam theory, which is suitable for slender beam configurations. In the current analysis, the beam possesses a length-to-thickness ratio of $\frac{l_p}{h_p} = 100$, indicating that the effects of transverse shear deformation and rotary inertia are negligible. Therefore, the Euler-Bernoulli beam model is considered appropriate for accurately predicting the vibration behavior of the proposed nanobeam system. The present formulation employs the Galerkin method with shifted Legendre polynomial admissible functions to solve the governing nonlocal differential equation of the perforated non-homogeneous nanobeam. Compared with several conventional numerical and semi-analytical approaches frequently adopted in nanobeam vibration analysis, such as the finite element method, differential quadrature method, finite difference method, Rayleigh-Ritz method, and mesh-based discretization techniques, the proposed formulation offers several important computational and mathematical advantages. One of the major strengths of shifted Legendre polynomials lies in their orthogonal nature, which significantly improves numerical stability and convergence characteristics. The orthogonality property minimizes numerical ill-conditioning and reduces computational errors during matrix formulation, particularly for higher-order vibration modes and strongly coupled nonlocal problems. Unlike mesh-dependent methods such as the finite element method, finite difference method, and differential quadrature techniques, the present semi-analytical Galerkin formulation avoids extensive spatial discretization and mesh-sensitivity issues. Consequently, it offers improved computational efficiency and robustness in analyzing structures with variable material properties, perforation distributions, and elastic foundation parameters. Therefore, the proposed formulation provides an accurate, computationally efficient, and numerically stable alternative for the vibration analysis of complex nanobeam systems encountered in advanced nano-engineering applications.

The transverse displacement is expressed as a finite polynomial expansion that satisfies the boundary conditions of the nanobeam. When this approximation is introduced into the governing equation, the problem reduces to a generalized eigenvalue form. This makes the analysis more straightforward while still giving accurate values of frequency parameters and the corresponding mode shapes.

To approximate the transverse displacement of the perforated nanobeam, the function $\bar{G}_p(X)$ is expressed in a finite series form following the approach in several studies [23–25]. The displacement field is written as:

$$\bar{G}_p(X) = \sum_{i=1}^{n} \mathrm{A}_i \boldsymbol{\Lambda}_i(X) \tag{18}$$

where, A_i denotes the unknown coefficients to be determined using the Legendre polynomials-based Galerkin method, and n denotes the number of terms considered in the approximation. The present work adopts shifted Legendre polynomials as the basis functions due to their inherent orthogonality and improved numerical behavior. Classical Legendre polynomials are defined over the interval $[-1, 1]$. Therefore, a linear mapping $\Theta = 2X - 1$ is introduced to transform the nanobeam domain $X \in [0, 1]$ into a standard interval. Accordingly, the basis functions are chosen as $\boldsymbol{\Lambda}_i(X) = P_i(2X - 1)$, where $P_i(\cdot)$ represents the Legendre polynomial of degree i.

4.1 Boundary Conditions

The perforated nanobeam with variable elastic foundation considered in this research is subjected to sliding end boundary conditions at both ends. Under this support configuration, the nanobeam is free to translate along the longitudinal direction while no rotational restraint or transverse shear reaction develops at the boundaries. As a result, both the slope of the displacement field and the corresponding shear force vanish at the two ends of the beam.

Mathematically, the sliding end boundary conditions can be expressed in terms of the non-dimensional displacement function $\bar{G}_p(X)$ as [26]:

$$\frac{d\bar{G}_p(0)}{dX} = 0, \frac{d\bar{G}_p(1)}{dX} = 0$$

This indicates that the slope of the deflection curve is zero at $X = 0$ and $X = 1$. In addition, the condition of zero shear force at both ends is written as [26]:

$$\frac{d}{dX}\left(E(X)I\frac{d^2\bar{G}_p(0)}{dX^2}\right) = 0, \frac{d}{dX}\left(E(X)I\frac{d^2\bar{G}_p(1)}{dX^2}\right) = 0$$

These boundary conditions impose four independent constraints on the assumed displacement function. Upon substituting them into the displacement function given in Eq. (18), the first four unknown coefficients are eliminated systematically. Consequently, the admissible form of the displacement function $\bar{G}_p(X)$ is reduced to a series containing $(n - 4)$ independent coefficients.

Thus, Eq. (18) is modified as:

$$\bar{G}_p(X) = \sum_{i=5}^{n} \mathrm{A}_i \overline{\boldsymbol{\Lambda}}_i(X) \tag{19}$$

where, $\overline{\boldsymbol{\Lambda}}_i(X)$ denotes the modified Legendre polynomials.

Residual corresponding to the dimensionless governing Eq. (17) is expressed as:

$$
\begin{aligned}
P\left(\bar{G}_p\right) = \frac{d^2}{dX^2}\left([EI]_{eq}^{*}\frac{d^2\bar{G}_p}{dX^2}\right) + \lambda^2\bar{\xi}^2\frac{d^2}{dX^2}\left(S_0^*\bar{G}_p - S_2^*\frac{h_p^2}{12\times l_p^2}\frac{d^2\bar{G}_p}{dX^2}\right) \\
+K_w^*\xi^2\frac{d^2}{dX^2}\left(\left(1+gX+hX^2\right)\bar{G}_p\right) - \lambda^2\left(S_0^*\bar{G}_p - S_2^*\frac{h_p^2}{12\times l_p^2}\frac{d^2\bar{G}_p}{dX^2}\right) + K_w^*\left(\left(1+gX+hX^2\right)\bar{G}_p\right)
\end{aligned}
\tag{20}
$$

Weight functions are taken as:

$$W_i = \overline{\mathbf{\Lambda}}_i(X), \quad i = 5, 6, \ldots, n \tag{21}$$

By taking the inner product of the residual with the chosen weight functions and setting it to zero, the following expression is obtained:

$$\left\langle P\left(\bar{G}_p\right), \bar{\Lambda}_i(X)\right\rangle = 0, i = 5, 6, \ldots, n \tag{22}$$

By inserting $\bar{G}_p(X)$ from Eq. (19) into the above Eq. (22), a reduced algebraic system with $(n-4)$ unknown coefficients is obtained. This system is then rearranged into the standard form of a generalized eigenvalue problem, enabling straightforward determination of the frequency parameter λ, as given below:

$$[S][A] = \lambda^2[\mathrm{M}][A] \tag{23}$$

where, $[S]$ and $[\mathrm{M}]$ represent the stiffness and mass matrix of the system, and unknown quantities are collected in vector form $[A] = [A_5, A_6, \ldots, A_n]^T$. Solving this problem yields a frequency parameter λ along with the corresponding set of coefficients. These coefficients are subsequently substituted into Eq. (19) to reconstruct vibration mode shapes of the non-homogeneous perforated nanobeam associated with each computed frequency.

5 Model Validation and Convergence Analysis

The numerical consistency of the formulation is first examined through a convergence study in which the first four frequency parameters are evaluated under sliding-end boundary conditions. The results in Table 1 show a clear and stable convergence trend, indicating that the computational approach is reliable. As this is the first research of its kind, frequency results for both sliding-end boundary conditions are not available.

Table 1. Convergence analysis of the first four frequency parameters of a non-homogeneous perforated nanobeam corresponding to $\eta = 0.6$, $H_n = 3$, $\alpha = \beta = 2$, $\gamma = \delta = 4$, $g = h = 5$, $\bar{\xi} = 0.2$ and $K_w^* = 10$

n	λ_1	λ_2	λ_3	λ_4
8	3.7017	8.0431	20.1212	—
9	3.7017	8.0432	20.1141	33.5642
10	3.7017	8.0432	20.1142	33.3960
11	3.7017	8.0432	20.1140	33.3967
12	3.7017	8.0432	20.1140	33.3933
13	3.7017	8.0432	20.1140	33.3934
14	3.7017	8.0432	20.1140	33.3935
15	3.7017	8.0432	20.1140	33.3935
16	3.7017	8.0432	20.1140	33.3935

Note: "—" indicates that no valid value was obtained.

Therefore, the obtained frequency parameters are validated in Tables 2 and 3 by comparing them with the benchmark results available [20, 27] under sliding-clamped and simply supported boundary conditions, respectively. In Table 2, the present model is reduced to a solid beam configuration by considering $\eta = 1$, $H_n = 0$, $\alpha = \beta = 0$, $\gamma = \delta = 0$, $g = h = 0$, $\bar{\xi} = 0$, and $K_w^* = 0$, and the corresponding relative errors are also evaluated. Furthermore, Table 3 presents the validation of the proposed formulation for a homogeneous perforated nanobeam under simply supported boundary conditions with the parameters values $\eta = 0.7$, $\bar{\xi} = 0.15, 0.25, 0.35$, $\alpha = \beta = 0$, $\gamma = \delta = 0$, $g = h = 0$, and $K_w^* = 0$. The excellent agreement between the present results and the reported literature confirms the accuracy and reliability of the proposed model. The close agreement confirms the accuracy of the model and demonstrates its ability to capture the dynamic behavior of perforated nanobeams on variable elastic foundations.

As the governing formulation is presented in a dimensionless framework, the results remain independent of any specific material selection. For this reason, no particular material properties are assigned to the nanobeam in the current investigation. The geometry [24] of the beam is characterized by a length of $l_p = 10,000$ nm, a width of $w_p = 1,000$ nm, and a thickness of $h_p = 100$ nm, which implies that the length-tothickness ratio is $\frac{l_p}{\mathrm{h}_p} = 100$. Parameters taken for the convergence analysis are $\eta = 0.6$, $H_n = 3$, $\alpha = \beta = 2$, $\gamma = \delta = 4$, $g = h = 5$, $\bar{\xi} = 0.2$ and $K_w^* = 10$.

Table 2. Comparison of the first four frequency parameters of a solid beam with those from existing literature under sliding-clamped boundary conditions

Frequencies (λ)	Present Results	Pakdemirli et al. [27]	Relative Error
λ_1	5.59318	5.59332	0.0000250
λ_2	30.21962	30.22580	0.0002045
λ_3	74.59787	74.63889	0.0005499
λ_4	138.64465	138.79125	0.0010574

Table 3. Comparison of the first three frequency parameters of a homogeneous perforated nanobeam with those from existing literature under simply supported boundary conditions

$\bar{\xi}$	H_n	η	λ_1		λ_2		λ_3	
			This work	Eltaher et al. [20]	This work	Eltaher et al. [20]	This work	Eltaher et al. [20]
0.15			9.00516	9.00516	28.97370	28.97370	51.71935	51.71935
0.25	2	0.7	7.82896	7.82896	21.38126	21.38126	34.98940	34.98940
0.35			6.69790	6.69790	16.48063	16.48063	25.98251	25.98251
0.15			8.82710	8.82710	28.40118	28.40118	50.69854	50.69854
0.25	4	0.7	7.67415	7.67415	20.95877	20.95877	34.29880	34.29880
0.35			6.56546	6.56546	16.15497	16.15497	25.46968	25.46968

6 Results and Discussion

The governing equations are reformulated through a linear transformation to facilitate this process. Furthermore, the strong dependence of perforation parameters, non-homogeneity parameters, and variable foundation parameters is briefly explained and clearly discussed.

Table 4 presents the first three frequency parameter values of a non-homogeneous perforated nanobeam which is rested on a variable elastic foundation with a sliding end boundary condition. For the tabulated values, the parameters $\gamma = 1$, $\delta = 2$, $g = 5$, $h = 10$, $\bar{\xi} = 0.1$ and $K_w^* = 5$ are considered, while η, H_n, α, and β are varying. It is worth mentioning that keeping η and H_n fixed, varying α and β leads to an increase in three frequency parameters. Also it can been seen the influence of H_n is modedependent.

Table 4. Frequency parameters of a perforated nanobeam with a variable elastic foundation under sliding-end boundary conditions with $\gamma = 1$, $\delta = 2$, $g = 5$, $h = 10$, $\bar{\xi} = 0.1$ and $K_w^* = 5$

η	H_n	α	β	λ_1	λ_2	λ_3
		1	1	4.9722	11.7183	37.3661
	1	5	5	4.9907	17.6155	58.0247
0.3		10	10	4.9957	22.5130	74.6701
		1	1	5.2488	10.2112	31.2519
	3	5	5	5.2818	14.9493	48.3845
		10	10	5.2905	18.9568	62.2124
		2	2	4.0716	11.5938	37.7676
	2	7	7	4.0807	16.9613	56.2204
0.7		12	12	4.0831	20.7454	68.9939
		2	2	4.0926	11.3946	37.0312
	4	7	7	4.1022	16.6456	55.1158
		12	12	4.1048	20.3510	67.6353

The variation in vibration frequency characteristics of a non-homogeneous perforated nanobeam under changes in material density parameters γ, δ and foundation parameter K_w^* is presented in Table 5. The first three frequency parameter values are shown in the table, with fixed parameters taken as $\alpha = 1$, $\beta = 2$, $g = 2$, $h = 5$, $\eta = 0.5$ and $H_n = 2$.

Table 5. Frequency parameters of a perforated nanobeam with a variable elastic foundation under sliding-end boundary conditions with $\alpha = 1$, $\beta = 2$, $g = 2$, $h = 5$, $\eta = 0.5$ and $H_n = 2$

$\bar{\xi}$	γ	δ	K_w^*	λ_1	λ_2	λ_3
		1	1	1.5671	10.8388	36.9404
			10	4.9296	11.7007	37.2118
0.1	1	5	1	1.2057	9.1577	30.1581
			10	3.8129	9.8371	30.3727
		10	1	0.9808	8.0527	25.7642
			10	3.0981	8.6586	25.9498
		2	5	2.7370	7.5202	17.4544
			15	4.7374	8.4570	17.9058
0.3	2	7	5	2.3718	6.5442	14.5655
			15	4.1074	7.3569	14.9412
		12	5	2.1207	5.8761	12.9652
			15	3.6687	6.6047	13.3133

From Table 5, it is observed that as the density parameter parameter δ increases from 1 to 10, all three frequency parameters decrease, since higher material density leads to lower natural frequencies. On the other hand, as the foundation parameter K_w^* increases, the frequencies increase due to the enhancement in system stiffness.

6.1 Effect of Perforation Parameters

The effect of perforation parameters, such as the filling ratio and the number of rows of holes, on the dynamical behavior, including vibration frequencies and corresponding mode shapes, is very crucial and important for engineering applications. Therefore, these effects need to be examined carefully. Figure 3 presents the influence of these perforation parameters on the frequencies and mode shapes.

From Figure 3a and 3b, it is observed that the first two frequency parameters exhibit distinct trends depending on the combination of filling ratio (η) and number of holes (H_n). For $H_n = 1, 2, 3, 4$, the frequency parameters consistently decrease with an increase in filling ratio under sliding-end boundary conditions. The reason is that increasing η adds more material to the nanobeam, which alters its stiffness and mass distribution, leading to more stable and convergent frequency behavior. Figure 3c shows the mode shape deflection of a non-homogeneous perforated nanobeam under sliding-end boundary conditions. It illustrates the deformation pattern of the beam during vibration, indicating how the displacement varies along its length.

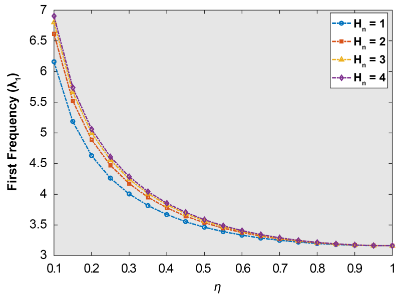
The figures are obtained using these fixed parameter values:
For frequency (λ): $\alpha = 1$, $\beta = 2$, $\gamma = 1$, $\delta = 2$, $g = 1$, $h = 2$, $\bar{\xi} = 0.1$ and $K_w^* = 10$.
For mode shape: $H_n = 3$, $\alpha = 5$, $\beta = 15$, $\gamma = 0$, $\delta = 0$, $g = 5$, $h = 15$, $\bar{\xi} = 0.3$ and $K_w^* = 50$.

6.2 Effect of Non-Homogeneity Parameters

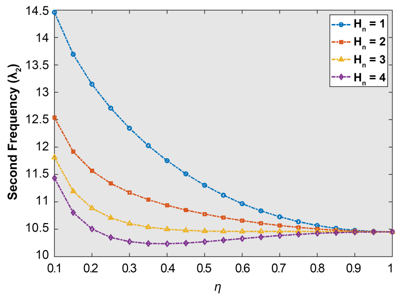
The effect of non-homogeneous material properties such as Young's modulus and material density parameters α, β, γ, and δ has a significant impact on the frequencies and mode shapes of a perforated nanobeam with a variable elastic foundation. Therefore, these effects must be investigated very carefully and rigorously. Figure 4a and 4b present first and second frequency parameters.

It can be observed that when material density parameters γ and δ increase and follow a quadratic variation, the frequency values decrease. This happens because an increase in material density increases the overall mass of the system, which in turn reduces the natural frequencies. In Figure 4a, when the filling ratio converges to 1, frequencies also decrease due to the presence of more material in the nanobeam. In Figure 4c, when α and β increase to 5, 10, 15, and 20, the first and second mode-shape profiles exhibit changes in curvature and spatial deformation patterns. This is because higher values of α and β enhance spatial variation of stiffness along the beam, which modifies the distribution of bending resistance. As a result, the mode shapes exhibit increased curvature and altered deformation patterns, rather than simply larger deformation amplitudes.

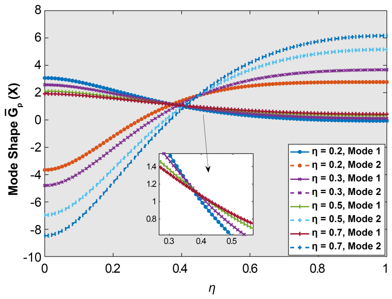
The results are obtained for the following fixed set of parameter values:
For frequency (λ): $H_n = 3$, $\alpha = 5$, $\beta = 20$, $g = 5$, $h = 20$, $\bar{\xi} = 0.2$ and $K_w^* = 10$.
For mode shape: $\eta = 0.7$, $H_n = 2$, $g = 5$, $h = 20$, $\gamma = 0$, $\delta = 0$, $\bar{\xi} = 0.3$ and $K_w^* = 50$.



(a)

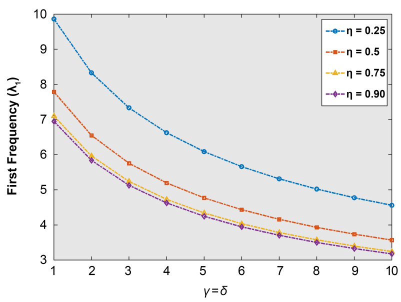


(b)

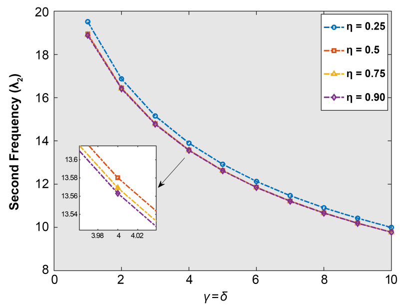


(c)

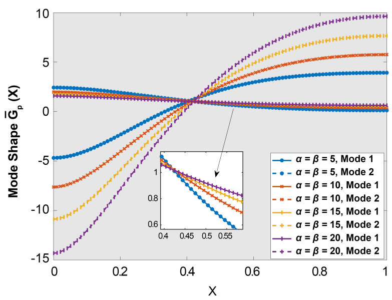
Figure 3. Frequency parameters and mode shape variation of a non-homogeneous perforated nanobeam with variations in filling ratio η and number of holes H_n: (a) variation of λ_1; (b) variation of λ_2; (c) first and second mode shapes



(a)



(b)



(c)

Figure 4. Frequency parameters and mode shape variation of a non-homogeneous perforated nanobeam with variations in non-homogeneous parameters α and β: (a) variation of λ_1 with γ and δ; (b) variation of λ_2 with γ and δ; (c) first and second mode shapes with different α and β

6.3 Effect of Variable Foundation

The dynamic behavior of a non-homogeneous perforated nanobeam is strongly governed by the characteristics of its supporting foundation. In this context, Figure 5a and 5b illustrate the influence of variable foundation parameters on the vibration response of the system.

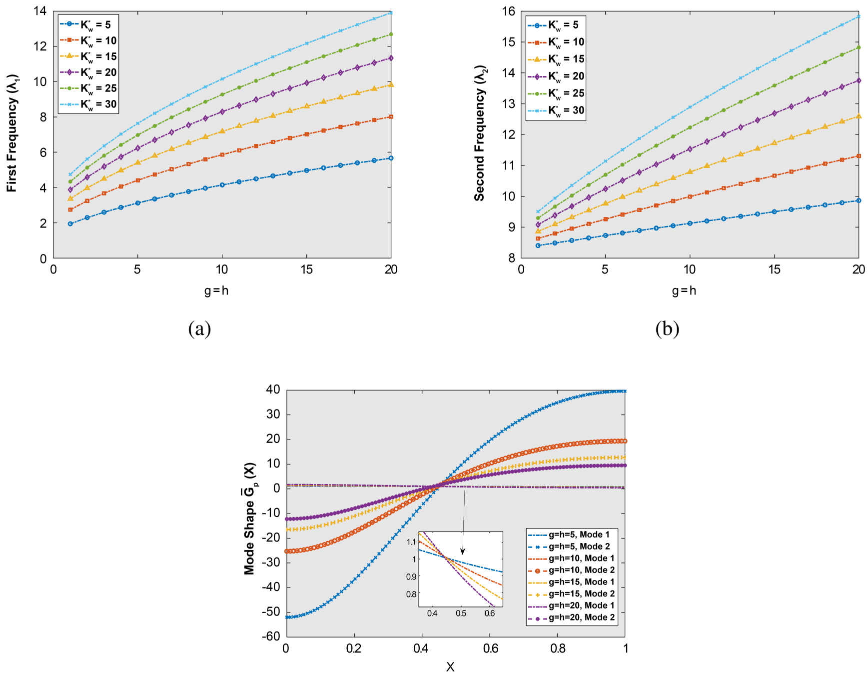
It can be observed that, as the values of the foundation parameters g and h increase, both the first and second frequency parameters increase for all considered values of K_w^*. This trend is physically reasonable. Higher values of g and h effectively enhance the stiffness of the elastic foundation, which provides greater resistance to deformation of the nanobeam. As a result, the overall system becomes stiffer, leading to a higher frequency parameter. Thus, the variation in foundation parameters directly modifies the stiffness characteristics of the system, and consequently has a strong influence on its vibrational response.

Figure 5c illustrates the first and second mode-shape profiles corresponding to their respective frequency parameters, highlighting the deformation characteristics of the nanobeam under free vibration. It is observed that variations in the foundation parameters g and h influence the spatial deformation patterns of both the first and second mode shapes. For the first mode, increasing the foundation parameters modifies the curvature of the mode-shape profile, indicating the influence of foundation stiffness on the beam deformation. A similar trend is observed for the second mode, where changes in g and h alter the deformation pattern and curvature of the mode shape. These observations demonstrate that the elastic foundation affects the distribution of deformation along the nanobeam, while the mode-shape profiles are presented using a consistent normalization procedure for comparison.

For the analysis of frequency parameters and mode shape, the following set of parameters is used:

For frequency (λ): $\eta = 0.8$, $H_n = 4$, $\alpha = 1$, $\beta = 5$, $\gamma = 1$, $\delta = 5$, and $\bar{\xi} = 0.3$.

For mode shape: $\eta = 0.8$, $H_n = 4$, $\alpha = 1$, $\beta = 5$, $\gamma = 0$, $\delta = 0$, $\bar{\xi} = 0.3$ and $K_w^* = 10$.



(a) (b) (c)

Figure 5. Frequency and mode shape variation of a non-homogeneous perforated nanobeam with respect to foundation parameters g and h: (a) variation of λ_1; (b) variation of λ_2; (c) first and second mode shapes

The coupled interaction between the nonlocal parameter, filling ratio, number of holes, density parameters, and foundation parameters significantly influences the dynamic response of the perforated nanobeam. As observed Figure 3a illustrates that the frequency parameters decrease with increasing filling ratio, Figure 4a indicates that increasing the non-homogeneous material parameters γ and δ decreases the frequency parameters due to the combined effects of stiffness degradation and mass redistribution along the beam. Conversely, Figure 5a shows that the frequency parameters increase with increasing foundation parameters g and h, since the enhanced elastic foundation stiffness provides greater structural support and increases the overall rigidity of the nanobeam. Similarly, variations in the number of holes (H_n) and material gradation parameters (α, β) alter the stiffness, mass distribution and mode shape characteristics of the system. At the nanoscale, these coupled effects become highly significant because small-scale interactions, material non-homogeneity, and support conditions collectively govern the vibrational behavior of nanostructures. Therefore, the present formulation provides a more realistic representation of nanobeam behavior under complex environmental and multiphysics operating conditions.

7 Conclusion

This research provides the vibrational response of non-homogeneous perforated nanobeams on variable elastic foundations under sliding-end boundary conditions. The results clearly quantify the coupled effects of material non-homogeneity, perforation characteristics, and foundation parameters on frequencies and corresponding mode shapes. It is demonstrated that when stiffness and mass density vary across the structure, they change how it behaves during vibration, which leads to noticeable differences in its frequencies. Furthermore, the inclusion of perforations modifies the effective structural stiffness, while variations in filling ratio and hole distribution directly influence the dynamic response. The role of foundation parameters is equally dominant, where an increase in foundation stiffness parameters enhances system rigidity and yields higher frequency values.

From a computational standpoint, the adopted formulation combining Euler-Bernoulli beam theory with Eringen's nonlocal elasticity and solved via the Galerkin approach using shifted Legendre polynomials exhibits strong numerical stability, rapid convergence, and high accuracy. The consistency of the obtained results with existing benchmarks validates the robustness of the proposed model. Overall, this investigation provides a reliable and efficient numerical framework capable of capturing the complex interplay between material gradation, perforation, and foundation variability. The outcomes offer direct applicability in the design of nanoscale beam-based devices where precise control of vibrational characteristics is critical. Future studies may extend the present model by incorporating thermoelastic, piezoelectric, fluid-coupled, and magneto-electro-elastic effects for advanced nanoscale applications. Additionally, nonlinear behavior, viscoelastic foundations, surface energy effects, and multiphysics coupling mechanisms may be considered to achieve a more realistic representation of nanoscale structural dynamics and smart nano-device performance. Therefore, this study provides a strong theoretical foundation for future developments in multifunctional coupled multiphysics nanostructural systems.

Author Contributions

Preliminary draft, procedure, results and validation, exploration, structured assessment, R.G. and A.K.G.; supervision, research design, idea, review and editing, S.C.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflict of Interest

The authors declare no conflicts of interest.

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Nomenclature

H_n	Number of rows of holes
η	Filling ratio
X	Axial position along the nanobeam length
l_p, h_p, w_p	Length, height and width of the nanobeam
α, β	Material non-homogeneity parameters
γ, δ	Density variation parameters
E	Young’s modulus
I	Moment of Inertia
ρ	Material density
A	Cross-sectional area
S_0	Spatial period
P_0	Period length
g, h	Foundation stiffness variation parameters
λ	Frequency parameter
$\bar{\xi}$	Non-local parameter
$\bar{G}_p(X)$	Transverse displacement
K_w^*	Foundation stiffness parameter
M_R	Bending moment resultant